

Flow

A High Performance Spark Data Processing Library

Chris Twiner

Copyright @ 2022 - UBS AG

Table of contents

1. Flow - 0.0.1-poc6	3
1.1 Process a Graph of data processing Steps with Quality	3

1. Flow - 0.0.1-poc6

Average			
Statement	96.39 %	Branch	90.74 %

1.1 Process a Graph of data processing Steps with Quality

Write auditable DQ and transformation rules using simple SQL or create re-usable functions via SQL Lambdas and manage them through a graph of Steps.

Your Flows can be provided DataFrames from either a Quality DataFrameLoader, catalog or even custom loading logic.

-  Initial Release

Entire Flows are lazily evaluated but support caching of a Steps results for interim performance benefits.

Each Step in a Flow has an associated Runner and, unless configured not to, keeps an audit of all the results of the previous Steps in the graph, either Quality:

1. engine,
2. collector,
3. folder,
4. DQ or
5. a custom implementation

The resulting Column then has a ResultApproach applied to it, either:

- AsIs - the audit and result column is added to the DataFrame,
- MergeFields - any nested result fields are merged with the DataFrame,
- OutputFieldsOnly - only the audit column and nested result columns are output,
- StarOnly - the audit column is dropped and only nested result columns are output,
- OutFieldOnly - only the result column remains or
- a custom implementation

Last update: May 3, 2026 12:33:20

Created: May 3, 2026 12:33:20